E.G.S. PILLAY ENGINEERING COLLEGE (Autonomous)

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Accredited by NBA T1 (B.E. – CIVIL, CSE, ECE, EEE, MECH & B.Tech – IT)
NAGAPATTINAM – 611002

B.E. ELECTRONICS AND COMMUNICATION ENGINEERING
R2023 - THIRD YEAR CURRICULUM

SEMESTER V

COURSE CODE	COURSE NAME	CATEGORY	L	T	P	C	MAX. MARKS		
							CA	ES	TOTAL
THEORY COURSES									
2302EC501	Digital Communication	PCC	3	0	0	3	40	60	100
2302EC502	Networks and Security	PCC	3	0	0	3	40	60	100
2302EC503	Transmission Lines and Waveguides	PCC	3	0	0	3	40	60	100
2302EC504	VLSI and Chip Design	PCC	3	0	0	3	40	60	100
	Professional Elective – I	PEC	3	0	0	3	40	60	100
	Open Elective – I	OEC	3	0	0	3	40	60	100
PRACTICAL COURSES									
2302EC551	Communication Systems Laboratory	PCC	0	0	2	1	60	40	100
2302EC552	VLSI and Chip Design Laboratory	PCC	0	0	2	1	60	40	100
OTHER COURSES									
2304GE501	Professional Development Course - III	EEC	0	0	2	1	100	--	100
2301LS501	Life Skills – V	LS	--	--	--	--	--	--	--
TOTAL						**21**	**460**	**440**	**900**

PROFESSIONAL ELECTIVE COURSES

Vertical I	Vertical II	Vertical III	Vertical IV	Vertical V	Vertical VI
Bio Medical Technologies	**Signal Processing**	**Semiconductor Chip Design and Testing**	**High Speed Communications**	**Space Technologies**	**Sensor Technologies and IoT**
Wearable Devices	Advanced Digital Signal Processing	Analog IC Design	4G/5G Communication Networks	Avionics Systems	Fundamentals of Nano-Electronics
Medical Imaging Systems	Computer Vision	Low Power IC Design	Advanced Wireless Communication Techniques	Positioning and Navigation Systems	Industrial IoT and Industry 4.0
Human Assist Devices	DSP Architecture and Programming	Mixed Signal IC Design Testing	Massive MIMO Networks	Radar Technologies	IoT Based System Design
Therapeutic Equipment	Image Processing	Validation and Testing Technology	Optical Communication & Networks	Remote Sensing	IoT Processors
Body Area Networks	Speech Processing	VLSI Testing and Design For Testability	Software Defined Networks	Rocketry and Space Mechanics	MEMS Design
Brain Computer Interface and Applications	Software Defined Radio	Wide Bandgap Devices	Wireless BroadBand Networks	Satellite Communication	Wireless Sensor Network Design

PROFESSIONAL ELECTIVE – I

S. NO.	COURSE CODE	COURSE TITLE	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
1.	2303EC001	Wearable Devices	PEC	3	0	0	3	3
2.	2303EC007	Advanced Digital Signal Processing	PEC	3	0	0	3	3
3.	2303EC013	Analog IC Design	PEC	3	0	0	3	3
4.	2303EC019	4G/5G Communication Networks	PEC	3	0	0	3	3
5.	2303EC025	Avionics Systems	PEC	3	0	0	3	3
6.	2303EC031	Fundamentals of Nano-Electronics	PEC	3	0	0	3	3

OPEN ELECTIVES – I

S.NO.	COURSE CODE	COURSE TITLE	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
1.	2303EC037	Analog and Digital Communication	OEC	3	0	0	3	3
2.	2303EC038	Display System	OEC	3	0	0	3	3
3.	2303EC039	Embedded Systems	OEC	3	0	0	3	3
4.	2303EC040	Generation of Communication Technology	OEC	3	0	0	3	3
5.	2303EC041	Robotics	OEC	3	0	0	3	3
6.	2303EC042	Measurement and Instrumentation	OEC	3	0	0	3	3

2302EC501	DIGITAL COMMUNICATION	L	T	P	C
		3	0	0	3

PREREQUISITE:

1. 2302EC301 - Signals and Systems
2. 2302EC402 - Analog Communication

COURSE OBJECTIVES:

- To understand the fundamentals of Pulse Modulation
- To analyze the Base band modulation
- To design passband modulation

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Describe the concepts of pulse modulation in communication systems
CO2:	Apply wave shaping techniques to minimize ISI
CO3:	Analyze and Design different types of passband modulation techniques
CO4:	Differentiate and apply Source Coding and Error control coding in Communication
CO5:	Analyze Spread spectrum Techniques

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	2							3
CO2	3	3	3	2							3
CO3	3	3	3	2							3
CO4	3	3	3	2							3
CO5	3	3	3	2							3

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1		3
CO2		3
CO3		3
CO4		3
CO5		3

COURSE CONTENTS:

MODULE I PULSE MODULATION 9 Hours

Block diagram of a Digital Communication systems - Sampling process (Qualitative): Low pass and Band pass sampling, Aliasing, Signal Reconstruction Quantization: Uniform & non uniform quantization, quantization noise, Logarithmic Companding of speech signal Waveform coding: Pulse Code Modulation (PCM), Differential pulse code modulation, Adaptive differential pulse code modulation, Delta modulation.

MODULE II BASEBAND MODULATION 7 Hours

Digital line encoding techniques: Need for line shaping of signals, Properties of Line codes, Unipolar / Polar RZ & NRZ, Bipolar NRZ, Manchester – Matched filter – Inter Symbol Interference and Nyquist criteria for ISI cancellation – Pulse shaping with raised cosine filter – Correlative level coding – M ary PAM transmission – Optimum linear receivers – Equalization techniques – Eye pattern.

MODULE III	**PASS BAND MODULATION**	**11 Hours**
Generation and detection of Amplitude Shift Keying (ASK) Modulation, Generation and detection of Frequency Shift Keying (FSK) Modulation, Generation and detection of Binary Phase Shift Keying (BPSK) Modulation, Generation and detection of Quaternary Phase Shift Keying QPSK) and QAM Performance of BPSK, QPSK and QAM in AWGN channel structure of Non-coherent Receivers, Principle of DPSK.		
MODULE IV	**ERROR CONTROL CODING**	**10 Hours**
Source Coding Theorem, Data Compaction (Lossless data compression - Huffman Coding, Prefix Coding), Discrete memory-less channel, Channel Capacity, Linear Block Codes (Syndrome decoding, Minimum Distance Considerations, Hamming Codes), Cyclic Codes, Convolutional Codes (Viterbi algorithm)		
MODULE V	**SPREAD SPECTRUM TECHNIQUES**	**8 Hours**
Spread Spectrum- PN Sequence code and properties – Direct Sequence and Frequency Hopping Spread Spectrum Systems –Processing gain and Jamming Margin.		
		TOTAL: 45 HOURS

REFERENCES:

1. *S. Haykin, Micheal Moher, Communication Systems, 5th Ed., John Wiley & Sons, 2021. 3 S. Haykin, Digital Communication Systems, John Wiley & Sons, 2013.*
2. *S. Haykin, Digital Communication Systems, John Wiley & Sons, 2013.*
3. *H. Taub and D. L. Schilling, Principles of Communication Systems, 4th edition, McGraw Hill, 2017.*
4. *George Kennedy and Bernard Davis," Electronic Communication systems", 6th Edition, TMH, 2017*
5. *https://moodle.najah.edu/mod/resource/view.php?id=48306*
6. *https://nptel.ac.in/courses/108102096/*
7. *https://nptel.ac.in/courses/Webcoursecontents/IIT%20Kharagpur/.../New_index1.html*
8. *https://ocw.mit.edu/.../lecture-videos/lecture-5-error-correction-syndromedecoding/*

2302EC502	NETWORKS AND SECURITY	L	T	P	C
		3	0	0	3

PREREQUISITE:

COURSE OBJECTIVES:

1. To gain knowledge on the various attacks in a network
2. To acquire knowledge on various encryption standards.
3. To build the ability to develop security standard based on the requirement.

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Identify vulnerability of computer networks to security threats.
CO2:	Acquire and understand the knowledge on existing security algorithms, cryptography standards, cryptography techniques and their implications on network security
CO3:	Understand various cyber-crimes and cyber security.
CO4:	Analyze the type of security threat and the appropriate security standard to be adopted
CO5:	Formulate and implement new security standards.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	2							3
CO2	3	3	3	2							3
CO3	3	3	3	2							3
CO4	3	3	3	2							3
CO5	3	3	3	2							3

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1		3
CO2		3
CO3		3
CO4		3
CO5		3

COURSE CONTENTS:

MODULE I	INTRODUCTION TO NETWORK SECURITY	9 Hours

Security Threats, Security Attacks, Security Services, Mechanisms - Model for Network Security - Classical Encryption Techniques – Substitutions - Transpositions Techniques - Stream Cipher, Block Cipher-Block Cipher Modes-ECB-CBC-CFB-OFB.

MODULE II	BLOCK CIPHERS AND THE DATA ENCRYPTION STANDARD,	9 Hours

Simple DES - Differential cryptanalysis - DES-Modes of operation - Triple DES-AES-RC4 – RSA. Hash Function-Message Digest algorithm (MD 5) - Secure Hash Algorithm - Diffie-Hellman Key Exchange - Key Management Techniques - Key Distribution - Key Agreement - Elliptic Curve Cryptography - Digital Signatures - Authentication Protocols.

MODULE III	CYBER CRIMES AND CYBER SECURITY	9 Hours

Cyber Crime and Information Security – classifications of Cyber Crimes – Tools and Methods – Password Cracking, Keyloggers, Spywares, SQL Injection – Network Access Control – Cloud Security – Web Security – Wireless Security		
MODULE IV	**SECURITY PRACTICE & SYSTEM SECURITY**	**9 Hours**
Authentication applications – Kerberos – X.509 Authentication services - Internet Firewalls for Trusted System: Roles of Firewalls – Firewall related terminology- Types of Firewalls - Firewall designs - SET for E-Commerce Transactions. Intruder – Intrusion detection system – Virus and related threats – Countermeasures – Firewalls design principles – Trusted systems – Practical implementation of cryptography and security.		
MODULE V	**E-MAIL, IP & WEB SECURITY**	**9 Hours**
E-mail Security: Security Services for E-mail-attacks possible through E-mail - establishing keys privacy - authentication of the source-Message Integrity-Non-repudiation-Pretty Good Privacy - S/MIME. IP Security: Overview of IPSec - IP and IPv6-Authentication Header-Encapsulation Security Payload (ESP) - Internet Key Exchange (Phases of IKE, ISAKMP/IKE Encoding). Web Security: SSL/TLS Basic Protocol - computing the keys - client authentication - PKI as deployed by SSL Attacks fixed in v3 – Exportability – Encoding - Secure Electronic Transaction (SET).		
		TOTAL: 45 HOURS

REFERENCES:

William Stallings, "Cryptography and Network Security: Principles and Practice", Prentice Hall Professional Technical Reference, Fourth Edition. 2004

Nina Godbole, Sunit Belapure, "Cyber Security: Understanding Cyber-crimes, Computer Forensics and Legal Perspectives", First Edition, Wiley India, 2011.

Alfred J. Menezes, Paul C.VanOorSchot, Scott A.Van Stone, "Handbook Of Applied Cryptography", CRC Press, 1996.

Atul Kahate, "Cryptography and Network Security", Tata McGraw-Hill

Bruce Schneier, "Applied Cryptography: Protocols, Algorithms, and Source Code in C", Second Edition.

Wiley, John & Sons, Incorporated, October 1995.

Richard E. Smith, "Internet Cryptography", Addison- Wesley, 1997.

2302EC503	TRANSMISSION LINES AND WAVEGUIDES	L	T	P	C
		3	0	0	3

PREREQUISITE:

1. 2302EC303 - Electromagnetic Fields

COURSE OBJECTIVES:

- To introduce the various types of transmission lines and its characteristics.
- To impart technical knowledge in impedance matching using Smith Chart.
- To illustrate the waveguide Structures propagation in TE, TM and TEM modes
- To understand the concepts of Rectangular and Circular waveguides
- To learn the concepts of a Planar transmission lines

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Explain the characteristics of transmission lines and its losses.
CO2:	Analyze impedance matching by stubs using Smith Charts.
CO3:	Analyze the electromagnetic fields configuration between the parallel plane and general wave behaviors along the uniform guiding structures.
CO4:	Comprehend the characteristics of TE and TM waves within a waveguide.
CO5:	Understand the basic characteristics of planar transmission line.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3		1			1			1
CO2	3	3	3		1			1			1
CO3	3	3	3		1			1			1
CO4	3	3	3		1			1			1
CO5	3	3	3		1			1			1

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1		2
CO2		2
CO3		2
CO4		2
CO5		2

COURSE CONTENTS:

MODULE I **TRANSMISSION LINE THEORY** **10 Hours**

General solution of transmission line – The two standard forms for voltage and current of a line terminated by an impedance – Physical significance of the equation and the infinite line – Wavelength and velocity of propagation – Waveform distortion – Distortion less transmission line – The telephone cable – Inductance loading of telephone cables – Input impedance of lossless lines – Reflection on a line not terminated by Z_O - Reflection coefficient - Reflection factor and reflection loss.

MODULE II **THE LINE AT RADIO FREQUENCIES** **10 Hours**

Standing waves and standing wave ratio on a line – One Eighth wave line – Quarter wave line and impedance matching – The half-wave line – Smith chart – Properties and Applications of the smith chart – Single stub

matching and double stub matching.		
MODULE III	**GUIDED WAVES**	**8 Hours**
Waves between parallel planes of perfect conductors – Transverse electric and transverse magnetic waves – Characteristics of TE and TM waves – Transverse electromagnetic waves – Attenuation of TE and TM waves in parallel plane guides –Wave impedances.		
MODULE IV	**RECTANGULAR AND CIRCULAR WAVEGUIDES**	**9 Hours**
Transverse magnetic waves in rectangular wave guides – Transverse electric waves in rectangular waveguides – Characteristics of TE and TM waves – Cutoff wavelength and phase velocity – Impossibility of TEM waves in waveguides – Dominant mode in rectangular waveguide – Attenuation of TE and TM modes in rectangular waveguide – Wave impedance – Characteristic impedance – Excitation of modes. Bessel functions – Solution of field equations in cylindrical co-ordinates – TM and TE waves in circular guides - Dominant mode in circular waveguide – Excitation of modes		
MODULE V	**PLANAR TRANSMISSION LINES**	**8 Hours**
Introduction to Planar transmission lines- Strip line – Microstrip Line: Basic Characteristics - Coplanar Waveguide & Coplanar Strip Line: Basic Characteristics, Slot Line: Basic Characteristics.		
		TOTAL: 45 HOURS

REFERENCES:

1. *Ryder J. D., "Networks, Lines and Fields", PHI, 2003.*
2. *Jordan E. C. and Balmain K. G., "Electro Magnetic Waves and Radiating System", PHI, 2003.*
3. *Ramo, Whineery and Van Duzer, "Fields and Waves in Communication Electronics", John Wiley, 2003.*
4. *David M. Pozar, "Microwave Engineering", 2nd Edition, John Wiley, 1997.*
5. *David K.Cheng, "Field and Waves in Electromagnetism", Pearson Education, 1989.*
6. *Anand K. Verma ,Introduction To Modern Planar Transmission Lines: Physical, Analytical, and Circuit Models Approach, Wiley – IEEE press , 2021*

2302EC504	VLSI AND CHIP DESIGN	L	T	P	C
		3	0	0	3

PREREQUISITE:

1. 2302EC302 – Digital Electronics

COURSE OBJECTIVES:

- To understand the fundamentals of CMOS fabrication process and CMOS circuits.
- To study and design combinational logic circuits using various Logic Styles.
- To provide basic knowledge about sequential logic circuits, memory, clocking and ASIC design.

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Demonstrate CMOS Fabrication process, MOS technology and Layout Design.
CO2:	Understand the Combinational logic circuits and design principles
CO3:	Understand the sequential circuits, CMOS Memories and clocking strategies.
CO4:	Describe the CMOS subsystem design process
CO5:	Understand the ASIC design process.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	3							3
CO2	3	3	3	3							3
CO3	3	3	3	3							3
CO4	3	3	3	3							3
CO5	3	3	3	3							3

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1	3	
CO2	3	
CO3	3	
CO4	3	
CO5	3	

COURSE CONTENTS:

MODULE I	**FABRICATION OF CMOS IC AND PHYSICAL DESIGN**	**9 Hours**

An overview of Silicon Semiconductor technology- NMOS fabrication - CMOS fabrication: n-well, P-well - Twin Tub and SOI Process - Layout design rules - Lambda Design Rules Stick Diagrams - VLSI Layout Design - Layout of Basic Structures - CMOS Logic Gates - Implementation of given logic function using CMOS logic.

MODULE II	**COMBINATIONAL LOGIC CIRCUITS AND DESIGN PROCESS**	**9 Hours**

Examples of combinational logic circuits, Pass Transistor and Transmission Gate, Static CMOS design, Pseudo NMOS –Dynamic CMOS logic, Clocked CMOS logic, Precharged domino logic- Keeper Circuits - Dual Rail - Cascode voltage Switch Logic, Low power design principles.

MODULE III	**SEQUENTIAL LOGIC CIRCUITS AND CLOCKING**	**9 Hours**

Sequencing Static Circuits, Conventional CMOS Latches and Flip-Flops, Klass Semi dynamic Flip-Flop (SDFF) – TSPC Latches and FF – Memory architecture- Flash Memory, CMOS Static RAM- Dynamic RAM and CAM - CMOS, Clocking Styles.		
MODULE IV	**ARITHMETIC CIRCUITS AND DIGITAL LOGIC DESIGN**	**9 Hours**
CMOS Mux - Equality Detector - Shift and Rotation Operation – Priority encoder - Ripple Carry Adder - Carry Look Ahead Adder - Carry Skip Adder - Carry select and Carry save - Adder – Braun / Baugh Wooley - Modified Booth, Encoded Multiplier.		
MODULE V	**ASIC DESIGN**	**9 Hours**
Introduction to ASIC design, Types of ASICS - Full Custom, Semi-Custom, Structured ASICs. Hardware Description Languages (HDLs) for ASIC Design, Introduction to physical design steps - Floor planning, Placement, and Routing Standard Cell Libraries and IP Integration.		
		TOTAL: 45 HOURS

REFERENCES:

1. *John P.Uyemura, "Introduction to VLSI circuits and systems", John Wiley & Sons, 2015.*
2. *Neil.H.E Weste, "Kamran Eshranghian principles of CMOS VLSI Design: A Systems Perspective", Addison Wesley, 2009.*
3. *Kamran Eshraghian, Douglas A. Pucknell, "Essentials of VLSI Circuits and Systems", Prentice Hall of India, 2015.*
4. *Keng, Lablebick, "CMOS Digital Integrated Circuits", Tata McGraw Hill, 2014.*
5. *Michael J smith, "Application specific integrated circuits", Addison Welsey.*

2302EC551	COMMUNICATION SYSTEMS LABORATORY	L	T	P	C
		0	0	2	1

PREREQUISITE:

1. 2302EC453 - Electronic Circuits Lab

COURSE OBJECTIVES:

To equip students with various issues related to analog and pulse communication
To design a Passband Modulation Techniques.
To generate different channel codes.

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Design and Analyze analog modulation Techniques.
CO2:	Design pulse modulation Techniques (PWM.PPM).
CO3:	Design and Analyze Passband modulation Techniques.
CO4:	Simulate Analog and Digital modulation Techniques.
CO5:	Simulate Error control and PN Sequence Generation Techniques.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	3				3		3	2
CO2	3	3	3	3				3		3	2
CO3	3	3	3	3				3		3	2
CO4	3	3	3	3	3			3		3	2
CO5	3	3	3	3	3			3		3	2

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1		3
CO2		3
CO3		3
CO4		3
CO5		3

LIST OF EXPERIMENTS:

1.	Amplitude Modulation and Demodulation	**2 Hours**
2.	Frequency Modulation and Demodulation	**2 Hours**
3.	Sampling and Reconstruction	**2 Hours**
4.	Time Division Multiplexing	**2 Hours**
5.	Pulse Code Modulation / Delta Modulation	**2 Hours**
6.	Generation of PWM/ PPM signals	**2 Hours**
7.	Passband Modulation and Demodulation : BPSK and BFSK	**2 Hours**
8.	Line Coding and Decoding Techniques	**2 Hours**
9.	Study of wireless Communication system using USRP/Deep Radio	**2 Hours**
Simulation using MATLAB		

1.	Analog Modulation	**2 Hours**
2.	Passband Digital Modulation	**2 Hours**
3.	Pre-emphasis and De-emphasis	**2 Hours**
4.	Frequency Division Multiplexing	**2 Hours**
5.	Error control Coding Techniques	**2 Hours**
6.	PN Sequence Generation	**2 Hours**
		TOTAL: 30 HOURS

REFERENCES:

6. *Simon Haykins, Communication Systems John Wiley & Sons 4th Edition, 2016.*
7. *K. Sam Shanmugam, Analog and Digital Communication, Wiley, 2005.*
8. *H. Taub and D. L. Schilling, Principles of Communication Systems, 4th edition, McGraw Hill, 2017.*
9. *George Kennedy and Bernard Davis," Electronic Communication systems", 6th Edition, TMH, 2017.*

2302EC552	VLSI AND CHIP DESIGN LABORATORY	L	T	P	C
		0	0	2	1

PREREQUISITE:

2302EC351 - Digital Electronics Laboratory

COURSE OBJECTIVES:

- To design, develop and simulate the digital circuits with Verilog HDL.
- To apply concepts and methods of Digital system design using HDL/FPGA.
- To understand the fundaments principles of VLS.I design in digital domain.
- To understand the fundaments principles of VLSI design in analog domain.
- To provide hands on design experience with EDA platforms.

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Write HDL code for basic as well as advanced digital integrated circuits.
CO2:	Import the logic modules into FPGA boards.
CO3:	Synthesize place and route the digital Intellectual properties.
CO4:	Design, simulate and extract the layouts of Digital & Analog IC blocks using EDA tools
CO5:	Test and verification IC design.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	3	3			3		3	3
CO2	3	3	3	3	3			3		3	3
CO3	3	3	3	3	3			3		3	3
CO4	3	3	3	3	3			3		3	3
CO5	3	3	3	3	3			3		3	3

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1	3	
CO2	3	
CO3	3	
CO4	3	
CO5	3	

LIST OF EXPERIMENTS:

1.	Design of basic combinational and sequential circuits using HDL. Simulate it using Xilinx/ Altera software.	**1. Hours**
2.	Design an adder; subtractor using HDL. Simulate it using Simulate it using Xilinx / Altera software.	**1. Hours**
2.	Design and implement shift register using HDL. Simulate it using Xilinx / Altera software.	**1. Hours**
2.	Design 3 bit synchronous up/down counter using HDL. Simulate it using Xilinx / Altera software and implement it by Xilinx/Altera FPGA	**1. Hours**
2.	Design 4 bit synchronous up/down counter using HDL. Simulate it using Xilinx /	**1. Hours**

	Altera software and implement it by Xilinx/Altera FPGA	
3.	Design and simulate CMOS basic gates; Flipflops. Generate Manual/ Automatic layout.	**2. Hours**
4.	Design 4 bit synchronous up/down counter using a Flipflop. Simulate it using Xilinx / Altera software and generate manual/ automatic layout.	**3. Hours**
5.	Design and simulate basic common source, common gate and common drain amplifiers.	**1. Hours**
2.	Design and simulate differential amplifier using EDA tool.	**1. Hours**
2.	Design and simulate CMOS inverting amplifier using EDA tool.	**3 Hours**
		TOTAL: 30 HOURS

REFERENCES:

10. *E. Fabricious, "Introduction to VLSI Design", 1st Edition, McGraw Hill, 2014.*
11. *Neil.H. E Weste David Harris, "CMOS VLSI Design: A Circuits and Systems Perspective", 4th Edition, Pearson Addison Wesley, 2015.*
12. *Kamran Eshraghian, Douglas A. Pucknell, "Essentials of VLSI Circuits and Systems", Prentice Hall of India, 2015.*
13. *John P.Uyemura, "Introduction to VLSI circuits and systems", John Wiley & Sons, 2015.*

PROFESSIONAL ELECTIVE – I

2303EC001	WEARABLE DEVICES	L	T	P	C
		3	0	0	3

PREREQUISITE:

COURSE OBJECTIVES:

- To know the hardware requirement of wearable systems
- To understand the communication and security aspects in the wearable devices
- To know the applications of wearable devices in the field of medicine

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Describe the concepts of wearable system.
CO2:	Explain the energy harvestings in wearable device.
CO3:	Use the concepts of BAN in health care.
CO4:	Illustrate the concept of smart textile
CO5:	Compare the various wearable devices in healthcare system

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3								3
CO2	3	3	3								3
CO3	3	3	3								3
CO4	3	3	3								3
CO5	3	3	3								3

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1	2	
CO2	2	
CO3	2	
CO4	2	
CO5	2	

COURSE CONTENTS:

MODULE I	INTRODUCTION TO WEARABLE SYSTEMS AND SENSORS	9 Hours

Wearable Systems- Introduction, Need for Wearable Systems, Drawbacks of Conventional Systems for Wearable Monitoring, Applications of Wearable Systems, Types of Wearable Systems, Components of wearable Systems. Sensors for wearable systems-Inertia movement sensors, Respiration activity sensor, Inductive Plethysmography, Impedance Plethysmography, Pneumography, Wearable ground reaction force sensor.

MODULE II	SIGNAL PROCESSING ANDENERGY HARVESTING FOR WEARABLEDEVICES	9 Hours

Wearability issues -physical shape and placement of sensor, Technical challenges - sensor design, signal acquisition, sampling frequency for reduced energy consumption, Rejection of irrelevant information. Power Requirements- Solar cell, Vibration based, Thermal based, Human body as a heat source for power generation, Hybrid thermoelectric photovoltaic energy harvests, Thermopiles.		
MODULE III	**WIRELESS HEALTH SYSTEMS**	**9 Hours**
Need for wireless monitoring, Definition of Body area network, BAN and Healthcare, Technical Challenges- System security and reliability, BAN Architecture – Introduction, Wireless communication Techniques.		
MODULE IV	**SMART TEXTILE**	**9 Hours**
Introduction to smart textile- Passive smart textile, active smart textile. Fabrication Techniques- Conductive Fibres, Treated Conductive Fibres, Conductive Fabrics, Conductive Inks. Case study- smart fabric for monitoring biological parameters - ECG, respiration		
MODULE V	**APPLICATIONS OF WEARABLE SYSTEMS**	**9 Hours**
Medical Diagnostics, Medical Monitoring-Patients with chronic disease, Hospital patients, Elderly patients, neural recording, Gait analysis, Sports Medicine.		
		TOTAL: 45 HOURS
REFERENCES:		
Sandeep K.S, Gupta, Tridib Mukherjee and Krishna Kumar Venkatasubramanian, Body Area Networks Safety, Security, and Sustainability, Cambridge University Press, 2013.		
Guang-Zhong Yang, Body Sensor Networks, Springer, 2006.		

2303EC007	ADVANCED DIGITAL SIGNAL PROCESSING	L	T	P	C
		3	0	0	3

PREREQUISITE:

1. 2302EC404 - Digital Signal Processing

COURSE OBJECTIVES:

- To introduce the concepts of discrete time random signal processing
- To know about multirate signal processing and its applications
- To understand the spectrum estimation techniques
- To learn the concept of prediction theory and filtering

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Comprehend multirate signal processing and demonstrate its applications
CO2:	Demonstrate an understanding of the power spectral density and apply to discrete random signals and systems
CO3:	Apply linear prediction and filtering techniques to discrete random signals for signal detection and estimation.
CO4:	Analyze adaptive filtering problems and demonstrate its application
CO5:	Apply power spectrum estimation techniques to random signals.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	2							1
CO2	3	3	3	2							2
CO3	3	3	3	2							2
CO4	3	3	3	2							2
CO5	3	3	2	2							1

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1		2
CO2		3
CO3		2
CO4		3
CO5		2

COURSE CONTENTS:

MODULE I	MULTIRATESIGNAL PROCESSING	9 Hours

Review of Convolution, DFT and ZT, Multirate Signal Processing - Decimation, Interpolation, Sampling Rate Conversion by a rational factor – digital filter banks, sub band coding, Quadrature Mirror Filter.

MODULE II	DISCRETE TIME RANDOM PROCESSES	9 Hours

Stationary random processes, Autocorrelation, Rational Power Spectra, Filters for generating random Processes from white noise and inverse filter – AR, MA and ARMA processes – relationship between autocorrelation and the filter parameters.

MODULE III	**LINEAR PREDICTION AND FILTERING**	**9 Hours**
Linear Prediction – Forward and Backward - Wiener filters for filtering and prediction – FIR Wiener Filter – IIR Wiener Filter – Kalman Filter.		
MODULE IV	**ADAPTIVE FILTERING**	**9 Hours**
FIR adaptive filters – adaptive filters based on steepest descent method – LMS algorithm – Variants of LMS algorithm – adaptive echo cancellation – adaptive channel equalization – RLS Algorithm.		
MODULE V	**SPECTRUM ESTIMATION**	**9 Hours**
Estimation of power spectra from finite duration observations of signals – Non parametric methods of spectrum estimation – the Bartlett and the Welch method – Parametric spectrum estimation – AR, MA and ARMA.		
		TOTAL: 45 HOURS

REFERENCES:

John G. Proakis & Dimitris G.Manolakis, ―Digital Signal Processing – Principles, Algorithms & Applications, Fourth Edition, Pearson Education / Prentice Hall, 2007.

P. Vaidyanathan, "Multirate systems and filter banks", Prentice Hall Inc. 1993.

Monson H. Hayes, "Statistical digital signal processing and modeling", John Wiley and Sons Inc. NewYork, Indian reprint 2008.

Haykin, Adaptive Filter Theory, 4thEdition, Pearson Education, New Delhi, 2006.

Sophoncles J. Orfanidis, "Optimum Signal Processing", McGraw Hill, 2000.

2303EC013	ANALOG IC DESIGN	L	T	P	C
		3	0	0	3

PREREQUISITE:

1. 2302EC401 Analog Integrated Circuits- Differential amplifiers, Application of operational amplifiers
2. 2302EC405 Electronic Circuits- Frequency response of amplifiers, Power amplifiers

COURSE OBJECTIVES:

- To know about mixed-signal devices and the need for testing these devices.
- To study the various techniques for testing.
- To learn about ADC and DAC based testing.
- To understand the Clock and Serial Data Communications Channels
- To study the general purpose measuring devices.

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Design amplifiers to meet user specifications.
CO2:	Analyse the frequency and noise performance of amplifiers.
CO3:	Design and analyse feedback amplifiers and one stage op amps .
CO4:	Analyse stability of op amp
CO5:	Testing experience of logic circuits.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	3	2	2		2		2	1
CO2	3	3	2	2	2	2		2		2	1
CO3	3	3	2	2	2	2		2		2	1
CO4	3	3	3	2	2	2		2		2	1
CO5	3	3	3	2	2	2		2		2	1

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1	3	
CO2	3	
CO3	3	
CO4	1	
CO5	2	

COURSE CONTENTS:

MODULE I	**SINGLE STAGE AMPLIFIERS**	**9 Hours**

Basic MOS physics and equivalent circuits and models, CS, CG and Source Follower, differential amplifier with active load, Cascode and Folded Cascode configurations with active load, design of Differential and Cascode Amplifiers – to meet specified SR, noise, gain, BW, ICMR and power dissipation, voltage swing, high gain amplifier structures.

MODULE II	**HIGH FREQUENCY AND NOISE CHARACTERISTICS OF AMPLIFIERS**	**9 Hours**

Miller effect, association of poles with nodes, frequency response of CS, CG and Source Follower, Cascode and Differential Amplifier stages, statistical characteristics of noise, noise in Single Stage amplifiers, noise in Differential Amplifiers.		
MODULE III	**FEEDBACK AND SINGLE STAGE OPERATIONAL AMPLIFIERS**	**9 Hours**
Properties and types of negative feedback circuits, effect of loading in feedback networks, operational amplifier performance parameters, single stage Op Amps, two-stage Op Amps, input range limitations, gain boosting, slew rate, power supply rejection, noise in Op Amps		
MODULE IV	**STABILITY, FREQUENCY COMPENSATION**	**9 Hours**
Multipole Systems, Phase Margin, Frequency Compensation, Compensation Of Two Stage Op Amps, Slewing In Two Stage Op Amps, Other Compensation Techniques.		
MODULE V	**LOGIC CIRCUIT TESTING**	**9 Hours**
Synchronous and Asynchronous Communications - Time-Domain Attributes of a Clock Signal - Frequency-Domain Attributes of a Clock Signal - Communicating Serially Over a Channel - Bit Error Rate Measurement - Methods to Speed Up BER Tests in Production - Deterministic Jitter Decomposition - Jitter Transmission Tests.		
		TOTAL: 45 HOURS

REFERENCES:

1. *Behzad Razavi, "Design of Analog CMOS Integrated Circuits", Tata Mcgraw Hill, 2001*
2. *Parag K.Lala, "An Introduction to Logic Circuit Testing", Morgan & Claypool Publishers, 2009*
3. *Willey M.C. Sansen, "Analog Design Essentials", Springer, 2006.*
4. *Grebene, "Bipolar and Mos Analog Integrated Circuit Design", John Wiley & Sons, Inc., 2003. Phillip E.Allen, Douglas R .Holberg, "CMOS Analog Circuit Design", Oxford University Press, 2nd Edition, 2002.*
5. *Recorded Lecture Available at http://www.ee.iitm.ac.in/vlsi/courses/ee5320_2021/start*
6. *Jacob Baker "CMOS: Circuit Design, Layout, And Simulation, Wiley IEEE Press, 3rd Edition,2010*

2303EC019	4G / 5G COMMUNICATION NETWORKS	L	T	P	C
		3	0	0	3

PREREQUISITE:

COURSE OBJECTIVES:

- To learn the evolution of wireless networks.
- To get acquainted with the fundamentals of 5G networks.
- To study the processes associated with 5G architecture.
- To study spectrum sharing and spectrum trading.
- To study spectrum sharing and spectrum trading.

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Describe the evolution of wireless networks.
CO2:	Explain the concepts of 5G networks.
CO3:	Comprehend the 5G architecture and protocols.
CO4:	Describe the dynamic spectrum management.
CO5:	Explain the security aspects in 5G networks.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
3	3	2	3	1					1	1	3
3	3	3	2	1					1	1	3
3	3	2	2	1					1	1	3
3	3	3	3	1					1	1	3
3	2	3	3	1					1	1	3

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1		1
CO2		1
CO3		2
CO4		2
CO5		2

COURSE CONTENTS:

MODULE I	**EVOLUTION OF WIRELESS NETWORKS**	**9 Hours**

Networks evolution: 2G,3G,4G, evolution of radio access networks, need for 5G. 4G versus 5G, Next Generation core(NG-core), visualized Evolved Packet core(vEPC).

MODULE II	**5G CONCEPTS AND CHALLENGES**	**9 Hours**

Fundamentals of 5G technologies, overview of 5G core network architecture,5G new radio and cloud technologies, Radio Access Technologies (RATs), EPC for 5G.

MODULE III	**NETWORK ARCHITECTURE AND THE PROCESSES**	**9 Hours**

5G architecture and core, network slicing, multi access edge computing(MEC)visualization of 5G components, end-to-end system architecture, service continuity, relation to EPC, and edge computing, 5G protocols: 5G NAS,NGAP, GTP-U, IPSec and GRE.		
MODULE IV	**DYNAMIC SPECTRUM MANAGEMENT AND MM-WAVES**	**9 Hours**
Mobility management, Command and control, spectrum sharing and spectrum trading, Cognitive Radio based on 5G, millimeter waves.		
MODULE V	**SECURITY IN 5G NETWORKS**	**9 Hours**
Security features in 5G networks, Network Domain Security, User Domain Security, Flow-based QoS framework, Mitigating the threats in 5G.		
		TOTAL: 45 HOURS
REFERENCES:		
5G Core networks: Powering Digitalization, Stephen Rommer, Academic Press, 2019.		
An Introduction to 5G Wireless Networks : Technology, Concepts and Use cases, Saro Velrajan, First Edition, 2020		
5G Simplified: ABCs of Advanced Mobile Communications Jyrki. T.J.Penttinen, Copyrighted Material.		
5G system Design: An end to end Perspective, Wan Lee Anthony, Springer Publications, 2019.		

2303EC025	AVIONICS SYSTEMS	L	T	P	C
		3	0	0	3

PREREQUISITE:

COURSE OBJECTIVES:

- To impart knowledge on the needs for avionics for both Civil and military aircraft.
- To impart knowledge on avionics architecture and Avionics data bus.
- To impart knowledge understand the various cockpit displays and human interfaces.
- To impart knowledge on the concepts of flight control systems, FMS and their importance.
- To impart knowledge on different navigation aids and need for certification

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Explain the different of Avionics Systems and its need for civil and military aircrafts considering the reliability and safety aspects
CO2:	Select a suitable architecture and data bus based on the requirements
CO3:	Compare the different display technologies used in cockpit
CO4:	Explain the principles of flight control systems and the importance of FMS
CO5:	Explain the communication and navigation techniques used in aircrafts

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	2	1				1		1
CO2	3	3	3	2	1				1		1
CO3	3	3	3	2	1				1		1
CO4	3	3	3	2	1				1		1
CO5	3	3	3	2	1				1		1

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1		2
CO2		2
CO3		2
CO4		2
CO5		3

COURSE CONTENTS:

MODULE I	**INTRODUCTION TO AVIONICS**	**9 Hours**

Basics of Avionics-Basics of Cockpits – Need for Avionics in civil and military aircraft and space systems – Integrated Avionics Architecture –Military and Civil system – Typical avionics System and Sub systems – Design and Technologies – Requirements and Importance of illities of Avionic Systems.

MODULE II	DIGITAL AVIONICS BUS ARCHITECTURE	**9 Hours**

Evolution of Avionics architecture– Avionics Data buses MIL-STD-1553, MIL-STD-1773, ARINC429, ARINC-629, AFDX/ARINC-664, ARINC-818 – Aircraft system Interface.

MODULE III	**COCKPIT DISPLAYS AND MAN-MACHINE INTERACTION**	**9 Hours**

Trends in display technology- CRT, LED, LCD, EL and plasma panel - Touch screen - Direct voice input (DVI) – Civil cockpit and military cockpit: MFD, MFK, HUD, HDD, HMD, HOTAS – Glass cockpit.		
MODULE IV	**FLIGHT CONTROL SYSTEMS**	**9 Hours**
Introduction to Flight control systems and FMS– Longitudinal control – Lateral Control –Autopilot – Flight planning – Radar Electronic Warfare - Certification-Military and civil aircrafts.		
MODULE V	**NAVIGATION SYSTEMS**	**9 Hours**
Overview of navigation systems - Communication Systems – Radio navigation – Types & Principles – Fundamentals of Inertial Sensors – INS – GNSS -- GPS – Approach and Landing Aids – ILS & MLS – Hybrid Navigation		
		TOTAL: 45 HOURS

REFERENCES:

R.P.G. Collinson, "Introduction to Avionics", Springer Publications, Third Edition, 2011.

Cary R .Spitzer, "The Avionics Handbook", CRC Press, 2000.

Middleton, D.H. "Avionics Systems", Longman Scientific and Technical, Longman Group UK Ltd., England, 1989.

Spitzer, C.R. "Digital Avionics Systems", Prentice Hall, Englewood Cliffs, N.J., U.S.A., 1987.

Myron Kayton, Walter R. Fried, "Avionics Navigation Systems", 2nd Edition, Wiley Publication, 2008.

Jim Curren, "Trend in Advanced Avionics", IOWA State University, 1992.

2303EC031	FUNDAMENTALS OF NANO-ELECTRONICS	L	T	P	C
		3	0	0	3

PREREQUISITE:

COURSE OBJECTIVES:

- To understand the concepts of Nano-Electronics and quantum electronics.
- To understand the concepts of Nano-Electronic devices, transistors, tunneling devices and superconducting devices.
- To understand the basics of Nanotube devices.

COURSE OUTCOMES:

On successful completion of this course, the student will be able to

CO1:	Understand the basics of Nano-Electronics including quantum wires, dots and wells.
CO2:	Use the mechanism behind Quantum Electronic devices.
CO3:	Understand the basics of Nano-Electronic transistors.
CO4:	Analyze the key performance aspects of tunneling and superconducting Nano-Electronic devices.
CO5:	Apply the knowledge in the development of Nanotubes and Nanostructure devices.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	2								2
CO2	3	3	3								2
CO3	3	3	3								2
CO4	3	3	2								2
CO5	3	3	3								2

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1	2	
CO2	2	
CO3	2	
CO4	2	
CO5	2	

COURSE CONTENTS:

MODULE I	INTRODUCTION TO NANO-ELECTRONICS	9 Hours

Scaling to Nano - Light as a wave and particle - Electrons as waves and particles - origin of quantum mechanics - General postulates of quantum mechanics - Time independent Schrodinger wave equation- Electron confinement - Quantum dots, wires and well-Spin and angular momentum

MODULE II	QUANTUM ELECTRONICS	9 Hours

Quantum electronic devices - Short channel MOS transistor - Split gate transistor - Electron wave transistor - Electron wave transistor - Electron spin transistor - Quantum cellular automata Quantum dot array, Quantum memory

MODULE III	NANO ELECTRONIC TRANSISTORS	9 Hours

Coulomb blockade - Coulomb blockade in Nano capacitors - Coulomb blockade in tunnel junctions - Single electron transistors, Semiconductor nanowire FETs and SETs, Molecular SETs and molecular electronics - Memory cell.		
MODULE IV	**NANO ELECTRONIC TUNNELING AND SUPER CONDUCTING DEVICES**	**9 Hours**
Tunnel effect - Tunneling element -Tunneling diode - Resonant tunneling diode - Three terminal resonant tunneling devices- Superconducting switching devices- Cryotron- Josephson tunneling device.		
MODULE V	**NANOTUBES AND NANOSTRUCTURE DEVICES**	**9 Hours**
Carbon Nanotube - Fullerenes - Types of nanotubes – Formation of Nanotubes –Assemblies – Purification of carbon Nanotubes – Electronic properties – Synthesis of carbon Nanotubes – Carbon Nanotube interconnects – Carbon Nanotube FETs and SETs –Nanotube for memory applications – Nano-structures and Nano-structured devices		
		TOTAL: 45 HOURS

REFERENCES

Hanson, "Fundamentals of Nanoelectronics", Pearson education, 2009.

Jan Dienstuhl, Karl Goser, and Peter Glosekotter, "Nanoelectronics and Nanosystems: From Transistors to Molecular and Quantum Devices", Springer-Verlag, 2004.

Mircea Dragoman and Daniela Dragoman, "Nanoelectronics: Principles and Devices", Artech House, 2009.

Robert Puers, Livio Baldi, Marcel Van de Voorde and Sebastiaan E. Van Nooten, "Nanoelectronics: Materials, Devices, Applications", Wiley, 2017

Brajesh Kumar Kaushik, "Nanoelectronics: Devices, Circuits and Systems", Elsevier science, 2018.

OPEN ELECTIVES – I

2303EC037	ANALOG AND DIGTIAL COMMUNICATION	L	T	P	C
		3	0	0	3

PREREQUISITE:

COURSE OBJECTIVES:

- To introduce the concepts of various modulations and their spectral characteristics.
- To learn Pulse modulation techniques.
- To understand the various Band pass signaling schemes and spread spectrum techniques.

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Examine the spectrum and methods of generation and detection of AM systems and its types.
CO2:	Develop the mathematical model for time domain representation, spectrum and methods of generation and detection of angle modulation systems.
CO3:	Apply the concepts of sampling process and determine the characteristics of Pulse Modulation schemes.
CO4:	Analyze the performance of different digital modulation /demodulation techniques
CO5:	Apply the knowledge on the principle of spread spectrum and synchronization.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	2								2
CO2	3	3	2								2
CO3	3	3	2								2
CO4	3	3	2								2
CO5	3	2	2								2

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1		2
CO2		2
CO3		2
CO4		2
CO5		2

COURSE CONTENTS:

MODULE I	AMPLITUDE MODULATION	9 Hours

Introduction to communication systems – Modulation – Need for modulation – Classifications of modulation techniques – Amplitude Modulation – Generation and Detection of AM – Transmitters and Receivers of AM –Super heterodyne receiver – Double Side Band Suppressed Carrier (DSBSC) systems - generation and detection – Single Side Band (SSB) systems – SSB-SC generation and detection, Vestigial Side Band (VSB) – Comparison of various AM systems.

MODULE II	ANGLE MODULATION	9 Hours

Frequency modulation: Narrowband and wideband FM – Generation of FM signal: Direct FM, indirect FM – Demodulation of FM signals using detectors – FM transmitters – FM receivers – Phase Modulation – Phase Locked Loop – Comparison of AM,FM and PM.		
MODULE III	**PULSE MODULATION TECHNIQUES**	**9 Hours**
PAM – PWM – PPM – Comparison of Pulse modulation – Sampling of Band limited signals – Anti aliasing and reconstruction filters - Quantization – Companding - Pulse Code Modulation – Differential pulse code modulation - Delta modulation – Adaptive Delta modulation – Intersymbol Interference – Eye pattern.		
MODULE IV	**PASSBAND DATA TRANSMISSION TECHNIQUES**	**9 Hours**
Generation, Detection, Representation of signal, Signal constellation diagram, Error probability and Power spectrum of ASK, FSK, BPSK, DPSK, QPSK, MSK, GMSK and QAM coherent schemes – Comparison and BER Analysis.		
MODULE V	**SYNCHRONIZATION AND SPREAD SPECTRUM TECHNIQUES**	**9 Hours**
Importance of Synchronization – Carrier, frame and symbol/Chip synchronization techniques, Spread Spectrum- PN Sequence code and properties – Direct Sequence and Frequency Hopping Spread Spectrum Systems –Processing gain and Jamming Margin – Multiple access techniques TDMA – FDMA – CDMA		
		TOTAL: 45 HOURS

REFERENCES:

4. *Simon Haykin, "Communication Systems" John Wiley &Sons, 4th Edition, 2016.*
5. *J.G. Proakis, "Digital Communications" McGraw Hill, 5th edition, 2007*
6. *B.P. Lathi, "Communication Systems" BS Publication-2004.*
7. *V.Chandrasekar, "Analog communication", Oxford University press, 2010*
8. *Bernard Sklar, "Digital Communication", 2nd Edition, Pearson Education, 2006.*
9. *Nptel link: https://nptel.ac.in/courses/117/105/117105143*
10. *Nptel link : https://onlinecourses.nptel.ac.in/noc20_ee17/course*

2303EC038	DISPLAY SYSTEM	L	T	P	C
		3	0	0	3

PREREQUISITE:

COURSE OBJECTIVES:

To expose the students to the basics of the display systems and to illustrate the current design practices of the display systems.

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Appreciate the technical requirement of different types of displays systems
CO2:	Analyze the various low power lighting systems
CO3:	Understand the operation of TFTs and LCD displays.
CO4:	Analyze the various kinds of emissive displays
CO5:	Critically evaluate the recent advancements in the displays device technology.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	2								2
CO2	3	2	2								2
CO3	3	2	2								2
CO4	3	2	2								2
CO5	3	2	2								2

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1	2	
CO2	2	
CO3	2	
CO4	2	
CO5	2	

COURSE CONTENTS:

MODULE I	**INTRODUCTION TO DISPLAY SYSTEMS**	**9 Hours**

Introduction to displays. Requirements of displays. Display technologies, CRT, Flat panel and advanced display technologies. Technical issues in displays.

MODULE II	**HEAD MOUNTED DISPLAY**	**9 Hours**

Head mounted displays. Displays less than and greater than 0.5 m diagonal. Low power and light emitting displays.

MODULE III	**WORKING OPERATION OF DISPLAY**	**9 Hours**

Operation of TFTs and MIMS. LCDs, Brightness. Types of LCD displays.

MODULE IV	**TYPES OF DISPLAY**	**9 Hours**

Emissive displays, ACTFEL, Plasma display and Field emission displays, operating principle and performance.

MODULE V	**APPLICATIONS OF DISPLAY**	**9 Hours**
Types of Displays: 3D, HDTV, LED, Touch screen.		
		TOTAL: 45 HOURS
REFERENCES:		
6. *L.W.Mackonald & A.C.Lowe, Display Systems, Design and Applications, Wiley, 2003.*		
7. *E.H.Stupp & M.S.Brennesholtz, Projection Displays, Wiley, 1999.*		
8. *PeterA.Keller, Electronic Display Measurement: Concepts, Techniques and Instrumentation, Wiley - Interscience, 1997.*		
9. *Recent literature in Display Systems.*		

2303EC039	EMBEDDED SYSTEM	L	T	P	C
		3	0	0	3

PREREQUISITE:

COURSE OBJECTIVES:

- Discuss the concepts of basic embedded systems.
- Describe about the ARM architecture.
- To introduce various technologies and protocols involved in Embedded communication protocols.
- To study about the different I/O device interfacing modules.
- To use the embedded controllers in real time applications.

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Outline the properties of embedded system.
CO2:	Point out the functionality of ARM processor
CO3:	Make use of the communication protocols in application specific purposes
CO4:	Interface I/O device peripherals with microcontroller
CO5:	Solve the real life problems using embedded systems

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	3								2
CO2	3	2	3								2
CO3	3	2	3								2
CO4	3	2	3								2
CO5	3	2	3								2

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1	3	
CO2	2	
CO3	2	
CO4	2	
CO5	3	

COURSE CONTENTS:

MODULE I	**INTRODUCTION**	**9 Hours**

Introduction to Embedded System, Embedded System Architecture, Embedded hardware, Embedded software, Classifications of Embedded Systems and Characteristics, Challenges and Design issues in Embedded systems, Embedded System on-chip.

MODULE II	**ARM PROCESSOR**	**9 Hours**

ARM processor naming, Types. CISC vs. RISC, Von-Neumann vs. Harvard architecture, ARM M3 features, Architecture, pipeline, Mode of operation, Instruction set, Exception handling.

MODULE III	**EMBEDDED COMMUNICATION PROTOCOLS**	**9 Hours**

Communication protocols – USART, I2C, CAN, SPI. Wireless communication protocols: Bluetooth, ZigBee, Z wave.		
MODULE IV	**I/O DEVICE INTERFACING**	**9 Hours**
C Programming, Interfacing Simple I/O Devices Like LED, Seven Segment, LCD, Switches, Motor (DC, Stepper, Servo), Relays and Sensors. Introduction to IOT.		
MODULE V	**EMBEDDED CONTROLLERS APPLICATION**	**9 Hours**
Home automation, Wireless sensor monitoring, Environmental monitoring, Gas leakage detection, Elevator design, Alarm clock using timers, Washing machine, Auto focusing Digital camera and Wearable devices.		
		TOTAL: 45 HOURS

REFERENCES:

1. *Raj Kamal, "Embedded Systems- Architecture, Programming and Design", Second Edition, Tata McGraw-Hill Publications, 2008.*
2. *Julio Sanchez Maria P.Canton, "Microcontroller Programming: The microchip PIC", CRC Press, Taylor & Francis Group, 2007.*
3. *The 8051 Microcontroller and Embedded Systems Using Assembly and C Second Edition Muhammad Ali Mazidi Janice Gillispie Mazidi Rolin D. McKinlay*
4. *Martin Bates, "Interfacing PIC Microcontrollers-Embedded Design by Interactive Simulation", Newnes Publication, 2006*

2303EC040	GENERATIONS OF COMMUNICATION TECHNOLOGY	L	T	P	C
		3	0	0	3

PREREQUISITE:

COURSE OBJECTIVES:

- To impart the fundamentals concepts of wireless communication systems.
- To introduce various technologies and protocols involved in wireless cellular communication.
- To differentiate the concepts of various generations in wireless cellular communication technology.

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Explain the evolution and concept of 1G communication technology
CO2:	Summarize the concept of 2G concept and architecture of cellular networks.
CO3:	Describe the 3G communication technology, concept and architecture.
CO4:	Elucidate the 4G networks and architecture
CO5:	Compare the different generations in communication.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	2	3								2
CO2	3	3	3								2
CO3	3	3	3								2
CO4	3	3	3								3
CO5	3	3	3								3

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1		2
CO2		3
CO3		3
CO4		3
CO5		3

COURSE CONTENTS:

MODULE I	1G EVOLUTIONS	9 Hours

History of Wireless Cellular Technology, Radio Communication, Concept of Cellular Radio System, Antenna used in 1G, Security measures in 1G, Advantages and Disadvantages in First Generation.

MODULE II	2G EVOLUTIONS	9 Hours

Review of cellular standards, Migration and Advancement of GSM Architecture and CDMA Architecture, WLAN – IEEE 802.11 and HIPERLAN, Bluetooth.

MODULE III	3G EVOLUTIONS	9 Hours

IMT-2000 - W-CDMA, CDMA 2000 – Radio & Network components, Network structure, Packet-data transport process flow, Channel Allocation, Core network, Interference-Mitigation techniques, UMTS-services, Air Interface, Network Architecture of 3GPP, UTRAN – Architecture, High Speed Packet Data-

HSDPA, HSUPA.		
MODULE IV	**4G EVOLUTION**	**9 Hours**
Introduction to LTE-A – Requirements and Challenges, Network architectures – EPC, E-UTRAN architecture - Mobility management, Resource management, Services, Channel-Logical and Transport channel mapping, Downlink/Uplink data transfer, MAC control element, PDU packet formats, Scheduling services, Random access procedure.		
MODULE V	**5G EVOLUTIONS**	**9 Hours**
Introduction, Need for 5G, Evolution of 5G, Comparison of different generations, QoS, 5G network architecture, Future enhancement.		
		TOTAL: 45 HOURS
REFERENCES:		

1. *Kaveh Pahlavan, "Principles of wireless networks", Prentice-Hall of India, 2008*
2. *A.F.Molisch, Wireless Communications, Wiley, 2005.*
3. *T.S.Rappaport, Wireless Communications: Principles and Practice, Second Edition, Pearson Education / Prentice Hall of India, Third Indian Reprint 2003.*
4. *Vijay K.Garg, "Wireless Network Evolution- 2G & 3G" Pearson, 2013.*
5. *K. Daniel Wong, "Fundamentals of Wireless Communication Engineering Technologies" Wiley, 2012.*
6. *P.Muthu Chidambara Nathan, Wireless Communications, PHI, 2008*
7. *A.Goldsmith, Wireless Communications, Cambridge University Press, 2005.*

2303EC041	ROBOTICS	L	T	P	C
		3	0	0	3

PREREQUISITE:

COURSE OBJECTIVES:

To demonstrate the concepts behind robots.

To interpret the electronics applications in robot for various purpose.

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1:	Understand the concepts behind robots.
CO2:	Articulate the concepts of drive systems in robots.
CO3:	Summarize the sensors used in robots with its purpose.
CO4:	Inspect various techniques related to sources and navigation of robot.
CO5:	Elaborate the applications of the robots.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	2								1
CO2	3	2	2								1
CO3	3	2	2								1
CO4	3	2	2								1
CO5	3	2	2								1

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1	1	
CO2	1	
CO3	1	
CO4	1	
CO5	1	

COURSE CONTENTS:

MODULE I	**INTRODUCTION**	**9 Hours**

Introduction – Definition and origin of robotics, Purpose of Robots, Artificial Intelligence, Robot Anatomy, Robot specifications, Robot characteristics – accuracy, precision, and repeatability, classification of robots, social issues of robotics.

MODULE II	**ROBOTIC DRIVE SYSTEMS**	**9 Hours**

Robotic drive systems and actuators: Hydraulic, Pneumatic and Electric drives. Specification, principle of operation and areas of application of: Stepper motor, Servo motor and brushless DC motor, Microprocessor control of electric motors, speed control using PWM and direction control using H- Bridge.

MODULE III	**SENSORS**	**9 Hours**

Position and displacement sensors, Strain gauge based force-torque sensors, Tachometers, Touch and Pressure, Piezoelectric material, Switches, Bend sensors, Pressure sensor, Smell, Humidity, Testing sensor.

MODULE IV	**POWER AND NAVIGATION**	**9 Hours**

Photovoltaic Cells, Fuel Cells, Li ion Batteries Vision, Voice communication, Route planning, Adaptive control, Error monitoring and recovery, Autonomy and intelligence in robots, Automated Guided Vehicles.		
MODULE V	**CASE STUDIES**	**9 Hours**
Pick and Place robot, Industrial applications of Robots in material handling and assembly, Speech-controlled mobile robot, Medical robots, Underwater bots, Aerobots, Drones And Robotic arm.		
		TOTAL: 45 HOURS
REFERENCES:		
Mikell and Groover, "Industrial Robotics – Technology, Programming and Applications", McGraw Hill, 2nd Edition, 2012.		
Saeed B. Niku, "Introduction to Robotics. Analysis and control, applications", Wiley student Edition, 2010.		
Spong and Vidyasagar, "Robot Dynamics and Control", John Wiley & Sons, 1990.		
D J Todd, "Fundamentals of Robot Technology", Springer Netherlands, 1986.		
Ashitava Ghosal, "Robotics, Fundamental concepts and analysis", Oxford University Press, 2006		

2303EC042	MEASUREMENTS AND INSTRUMENTATION	L	T	P	C
		3	0	0	3

PREREQUISITE:

COURSE OBJECTIVES:

- Learn the use of DC and AC bridges for measuring R, L and C.
- To introduce the concepts of Motion and Ranging Sensors.
- To study about the knowledge such as Optical, Pressure and Temperature Sensors.
- Discuss about the different types of Function generator and Wave analyzer.
- Learn the use of different types of displays, Electromagnetic and electrostatic interference.

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

CO1: Design different Bridge configurations and their applications.
CO2: Articulate different types of Ranging sensors.
CO3: Summarize different types of measuring sensors.
CO4: Outline the different type of function generators.
CO5: Understand different types of displays.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	3								1
CO2	3	2	3								1
CO3	3	2	3								1
CO4	3	2	3								1
CO5	3	2	3								1

COs Vs PSOs MAPPING:

COs	PSO1	PSO2
CO1	2	
CO2	2	
CO3	2	
CO4	2	
CO5	2	

COURSE CONTENTS:

MODULE I	MEASUREMENT CONCEPTS	9 Hours

Principles of operation and construction of PMMC-Static and dynamic characteristics-units and standards of measurements - error analysis - moving coil, moving iron meters, multi meters - True RMS Meters - Bridge measurements: Maxwell, Kelvin, Hay, Schering, Anderson and Wien bridge - Q meters.

MODULE II	MOTION, PROXIMITY AND RANGING SENSORS	9 Hours

Motion Sensors – Potentiometers, Encoders – Optical, Magnetic, Inductive, Capacitive, LVDT – RVDT, Accelerometer – GPS, Bluetooth, Range Sensors – RF beacons, Ultrasonic Ranging, Reflective beacons,

Laser Range Sensor (LIDAR).		
MODULE III	**OPTICAL, PRESSURE AND TEMPERATURE SENSORS**	**9 Hours**
Photo conductive cell, Photo voltaic, Photo resistive, LDR – Fibre optic sensors – Pressure – Diaphragm, Bellows, Piezoelectric – Tactile sensors, Temperature – IC, Thermistor, RTD, Thermocouple. Acoustic Sensors – flow and level measurement, Radiation Sensors - Smart Sensors - Film sensor, MEMS & Nano Sensors, LASER sensors.		
MODULE IV	**FUNCTION GENERATORS**	**9 Hours**
Function generators - RF signal generators - Sweep generators - Frequency synthesizer - Wave analyzer - Harmonic distortion analyzer - Spectrum analyzer - Heterodyne wave analyzer - Frequency counters - Time Interval measurement - Measurement of voltage, current, phase and frequency using CRO.		
MODULE V	**DISPLAY AND RECORDING DEVICES**	**9 Hours**
LED & LCD Display Dot Matrix Display, 7 Segment Display Strip Chart Recorders Single point and multi point Recorders – X-Y Recorders - Magnetic Tape Recorders - Data Loggers – Electromagnetic and Electrostatic interference.		
		TOTAL: 45 HOURS
REFERENCES:		

1. *Ernest, Doeblin, Dhanesh and N.Manik, "Measurement Systems - Application and Design", Tata McGraw-Hill, 2009.*
2. *Sawney A K and Puneet Sawney, "A Course in Mechanical Measurements and Instrumentation and Control", 12th Edition, Dhanpat Rai & Co, New Delhi, 2013.*
3. *Albert D.Helfrick and William D.Cooper, "Modern Electronic Instrumentation and Measurement Techniques", PHI, 2003.*
4. *B.C.Nakara, K.K.Chaudhry, "Instrumentation Measurement and Analysis", Tata McGraw-Hill, 2004.*
5. *Joseph J.Carr, "Elements of Electronics Instrumentation and Measurement", PHI, 2003.*
6. *Alan. S. Morris, "Principles of Measurements and Instrumentation", PHI, 2003.*